

# AMAR DHILLON

NYC Area | [amardhillon844@gmail.com](mailto:amardhillon844@gmail.com) | (XXX) XXX-XXXX | [linkedin.com/in/amar-dhillon-917537261](https://linkedin.com/in/amar-dhillon-917537261) | [github.com/AmarDhillon05](https://github.com/AmarDhillon05)

## Education

### University of Maryland, College Park

*B.S. in Computer Science, Minor in Statistics; GPA: 3.5*

Expected May 2028

*College Park, MD*

## Projects

### IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Designed the book as std::list FIFOs per price level inside a std::map with an order-ID hash map to make cancels O(1), which sustained **1.26M events/sec** over the full replay and a 1.2 μs per-message p99 on a separate 2M-event run
- Implemented B+ tree and LSM-tree backends for the engine's event log to compare them on identical 2M-record workloads, which showed the B+ tree giving **12.8x** the read throughput and 22x faster range queries for 27% more disk
- Traced an apparent 1.65x regression to the CPU frequency governor with perf counters and a do-nothing control book to rule out a code change, which led to core pinning that cut run-to-run variance from 6% to 0.06%

## Experience

### Amazon Web Services

Sep 2026 – Dec 2026

*Software Development Engineer Intern*

*Redmond, WA*

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

### Capital One

Jun 2026 – Aug 2026

*Software Engineering Intern*

*Richmond, VA*

- Cut AI workflow generation time **65%** to under 15s by splitting a Claude skill into focused sub-skills that call Python search and validation tools instead of searching open-ended
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits, which became the Claude skill's main search tool

### UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

*Software Engineer (Contract)*

*College Park, MD*

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

### UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

*Machine Learning Research Intern*

*College Park, MD*

- Built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D to let a vision-language model pick robot grasps for a PhD-led ICLR project, and tuned its prompts to favor grasps that avoid hitting the table

### TerpLabs

Jun 2025 – Present

*Co-Founder, Software Engineer*

*College Park, MD*

- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

## Technical Skills

**Languages:** C++17, C, Python, Java, Rust, SQL, TypeScript, Bash

**Systems & Performance:** debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking

**Cloud & Tools:** AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, unit testing (Pytest)

**Certification:** AWS Certified Developer – Associate (Nov 2024 – Nov 2027)